

refined, and combined into powerful trading systems.

Functions are to programming what standard operating procedures are to business. They define a clear process for accomplishing specific tasks, which can then be executed whenever needed without rewriting the instructions. Just as a chef doesn't reinvent their signature recipe each time they prepare it, programmers don't rewrite common operations every time they need them.

In the world of algorithmic trading, functions are especially valuable because they:

1.
Encapsulate trading logic — Package complex calculations into easy-to-use components
2.
Enable strategy testing — Allow traders to evaluate ideas across different market conditions
3.
Improve reliability — Reduce errors by defining operations once and reusing them
4.
Facilitate collaboration — Create a shared vocabulary for teams building trading systems
- 5.

In this chapter, we'll explore how to create and use functions in Python, and how they form the foundation of sophisticated trading systems. By the end, you'll be able to build your own reusable code components—the first step toward developing a professional-grade algorithmic trading system.

Defining Functions: Creating Your Trading Toolkit

Think of creating a function like establishing a new trading desk at an investment firm. First, you need to define its purpose and operations. Then, you staff it with the right people and resources. Finally, you set up the processes for how information flows in and out.